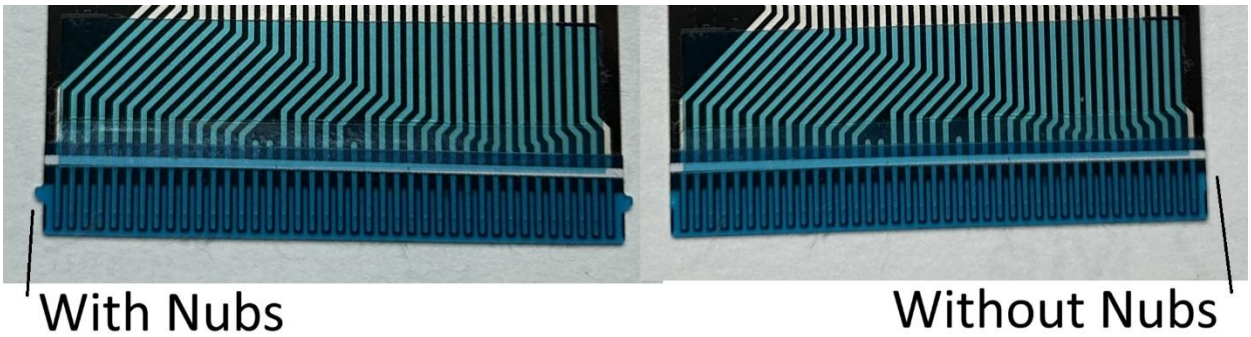
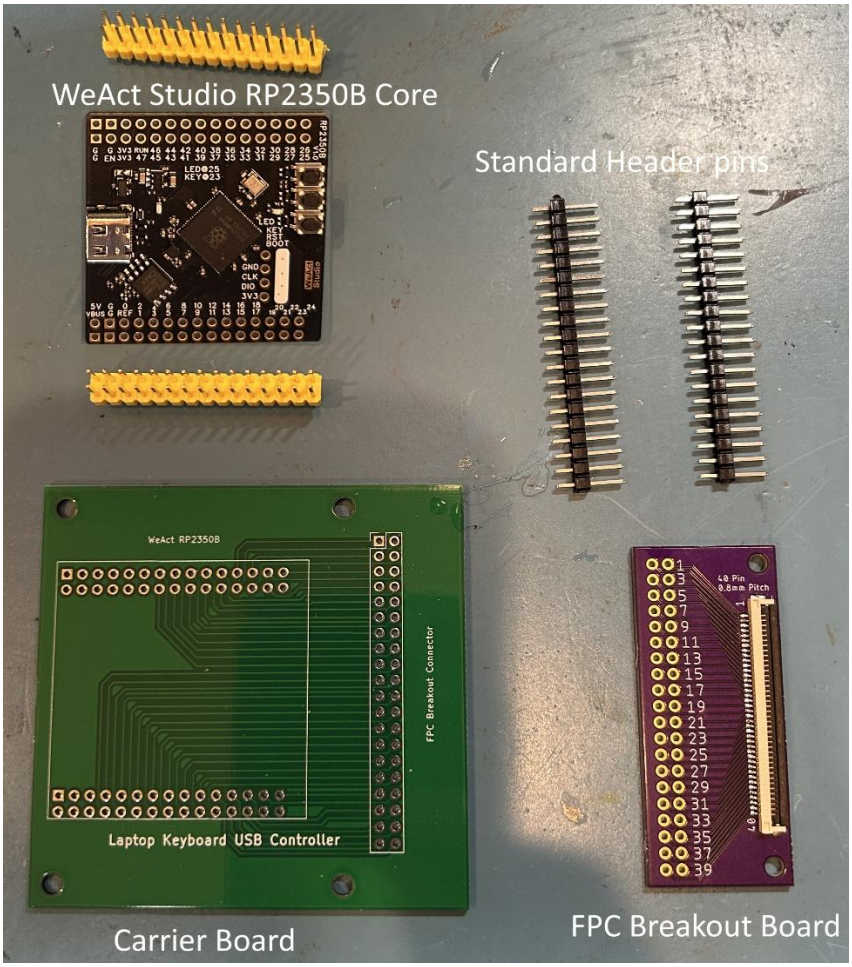
For this project, a [WeAct Studio RP2350B Core]() is attached to an HP Envy X360 16 inch laptop keyboard.

There were locking nubs on the cable that I trimmed off with scissors. This allowed me to use a generic FPC connector.



I designed the carrier board to connect 40 GPIO’s from the WeAct Studio RP2350B Core to a standard FPC breakout board. The KiCad carrier board file “WeAct2350_KB_Controller.kicad_pcb” and the zipped Gerber file for manufacturing are at my repo.



This particular keyboard has an FPC cable with 40 pins and a pitch of 0.8mm so a compatible breakout board was used. For keyboards with less pins, install the FPC breakout board where ever is convenient. The software will determine which keyboard keys are attached to the GPIO pins. Note that GPIO's 0, 23, 25, 43-47 are not used by the carrier board.

The following table gives the FPC to GPIO pin connections.

FPC Pin Numbers	WeAct RP2350B GPIO
1	32
2	31
3	30
4	29
5	26
6	28
7	27
8	34
9	33
10	36
11	35
12	38
13	37
14	40
15	39
16	42
17	41
18	1
19	2
20	3
21	4
22	5
23	6
24	7
25	8
26	9
27	10
28	11
29	12
30	13
31	14
32	15
33	24
34	22
35	21
36	20
37	19
38	18
39	17
40	16

This project uses the same basic software and procedure for loading the libraries as described in my Solderparty Stamp XL documentation. The WeAct Studio RP2350B will not erase the flash memory when loaded with the flash_nuke.uf2 program. I suspect I need to find a newer flash_nuke but have not investigated further. The WeAct Studio RP2350B Core was recently added to the Adafruit Circuit Python page. Use the latest Circuit Python uf2 file from here. I created a new version of my matrix decoder Python code to account for the additional GPIO's. It is a little squirrely to get to run, probably just my Python inexperience. It eventually gave me pin numbers for all the keys (see below) but it had I/O 3 as a row and a column so I had to do some editing.

GPIO connection list for the HP Envy X360 16 inch laptop keyboard.

KC.LCTRL	24	22
KC.RCTRL	24	21
KC.LSHIFT	20	18
KC.RSHIFT	18	16
KC.LALT	19	5
KC.RALT	13	5
KC.LGUI	17	7
KC.RGUI		
FN	14	8
KC.A	21	12
KC.B	13	11
KC.C	20	10
KC.D	21	10
KC.E	16	10
KC.F	21	11
KC.G	17	11
KC.H	17	9
KC.I	16	6
KC.J	21	9
KC.K	21	6
KC.L	21	3
KC.M	20	9
KC.N	13	9
KC.O	16	3
KC.P	41	16

KC.Q	16	12
KC.R	16	11
KC.S	21	15
KC.T	22	11
KC.U	16	9
KC.V	20	11
KC.W	16	15
KC.X	20	15
KC.Y	22	9
KC.Z	20	12
KC.GRAVE	14	12
KC.N1	19	12
KC.N2	19	15
KC.N3	19	10
KC.N4	19	11
KC.N5	14	11
KC.N6	14	9
KC.N7	19	9
KC.N8	19	6
KC.N9	19	3
KC.N0	41	19
KC.MINUS	41	14
KC.EQUAL	41	13
KC.BSPACE	17	2
KC.ESC	17	12
KC.F1	13	10
KC.F2	14	10
KC.F3	22	10
KC.F4	17	10
KC.F5	15	14
KC.F6	22	15
KC.F7	17	15
KC.F8	15	13
KC.F9	14	3
KC.F10	22	3

KC.F11		17	3
KC.F12		13	3
KC.INSERT			
KC.DELETE		13	1
KC.RIGHT		19	1
KC.LEFT		16	1
KC.UP		16	2
KC.DOWN		19	2
KC.SLASH		41	20
KC.DOT		20	3
KC.COMMA		20	6
KC.SCOLON		41	21
KC.QUOTE		41	17
KC.ENTER		21	2
KC.LBRC		41	22
KC.RBRC		22	6
KC.BSLASH		14	2
KC.CAPS		13	12
KC.TAB		22	12
KC.SPACE		20	2
KC.HOME		22	1
KC.END		20	1
KC.PGUP		14	1
KC.PGDOWN		21	1
KC.AUDIO_MUTE	Func	15	14
KC.AUDIO_VOL_UP	Func	17	15
KC.AUDIO_VOL_DOWN	Func	22	15
KC.BRIGHTNESS_UP	Func	22	10
KC.BRIGHTNESS_DOWN	Func	14	10
KC.MEDIA_NEXT_TRACK	Func	17	3
KC.MEDIA_PREV_TRACK	Func	14	3

The Normal key matrix.

GPIO#	13	14	16	17	19	20	21	22
1	DEL	PGUP	LEFT		RIGHT	END	PGDOWN	HOME
2		BSLASH	UP	BSPACE	DOWN	SPACE	ENTER	
3	F12	F9	O	F11	N9	DOT	L	F10
5	RALT				LALT			
6			I		N8	COMMA	K	RBRC
7				LGUI				
8		FN						
9	N	N6	U	H	N7	M	J	Y
10	F1	F2	E	F4	N3	C	D	F3
11	B	N5	R	G	N4	V	F	T
12	CAPS	GRAVE	Q	ESC	N1	Z	A	TAB
15	F8	F5	W	F7	N2	X	S	F6
18			RSHIFT			LSHIFT		
24							RCTRL	LCTRL
41	EQUAL	MINUS	P	QUOTE	N0	SLASH	SCOLON	LBRC

The Media matrix is the same as above except with the desired Fn-Media keys (shown in bold).

	13	14	16	17	19	20	21	22
1	DEL	PGUP	LEFT		RIGHT	END	PGDOWN	HOME
2		BSLASH	UP	BSPACE	DOWN	SPACE	ENTER	
3	F12	**MPRV**	O	**MNXT**	N9	DOT	L	F10
5	RALT				LALT			
6			I		N8	COMMA	K	RBRC
7				LGUI				
8		FN						
9	N	N6	U	H	N7	M	J	Y
10	F1	BRID	E	F4	N3	C	D	**BRIU**
11	B	N5	R	G	N4	V	F	T
12	CAPS	GRAVE	Q	ESC	N1	Z	A	TAB
15	F8	**MUTE**	W	**VOLU**	N2	X	S	**VOLD**
18			RSHIFT			LSHIFT		
24							RCTRL	LCTRL
41	EQUAL	MINUS	P	QUOTE	N0	SLASH	SCOLON	LBRC

I modified Marcel's Python code to convert the connection list into a normal key matrix. I also made the matrix manually to check the program and they were the same. The matrixes were put into KMK and worked fine with no odd behaviors. As the WeAct module gets more users and the libraries and documentation become more mature, it should be a popular way to make a laptop keyboard controller thanks to its low price and high I/O count.